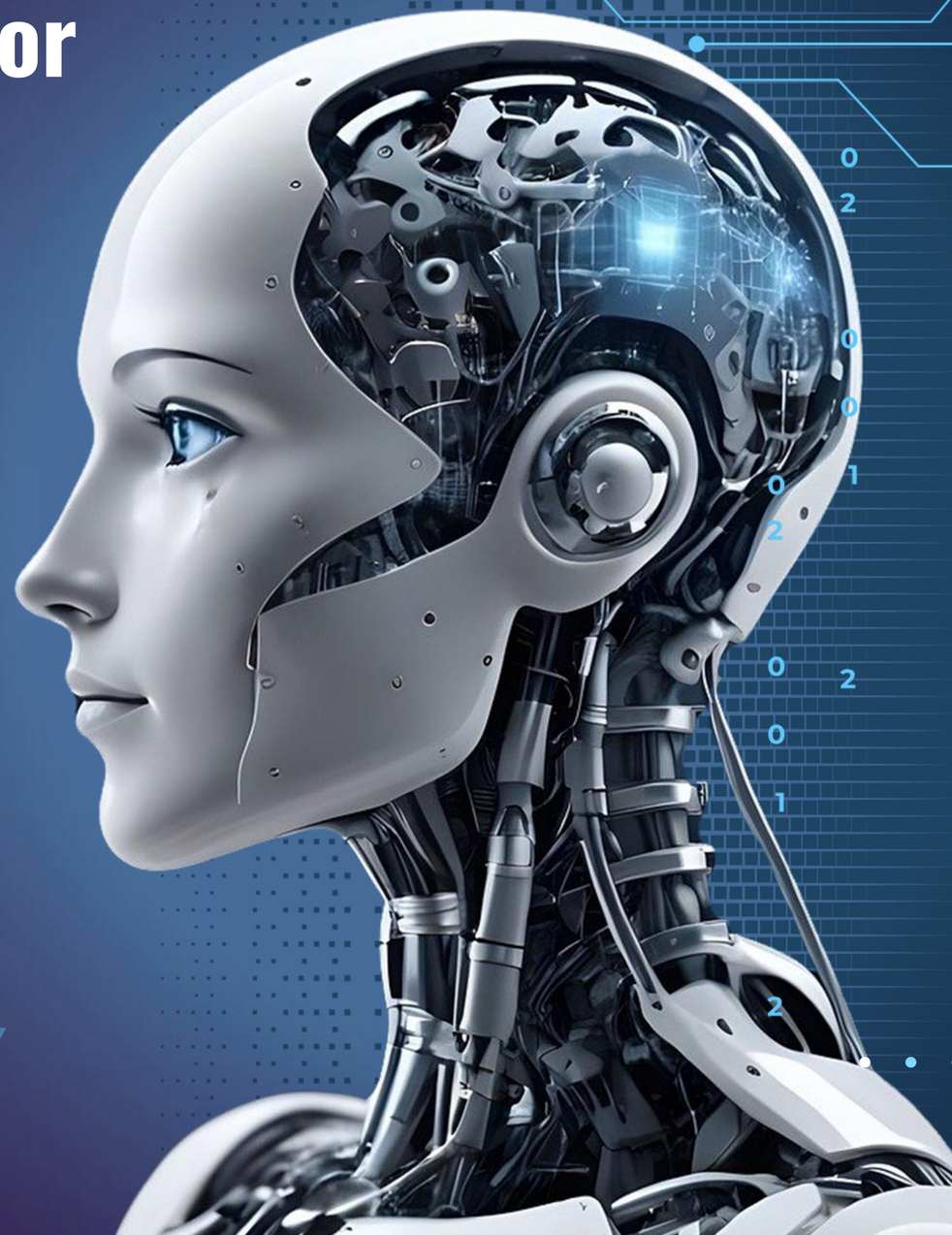
Multimodal Acoustic Feature Analysis for Parkinson's Detection Using Spectrograms and Formants with Deep Learning & Machine Learning Models

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Presentation Outline

01. Background & Motivation

02. Dataset (PC-GITA)

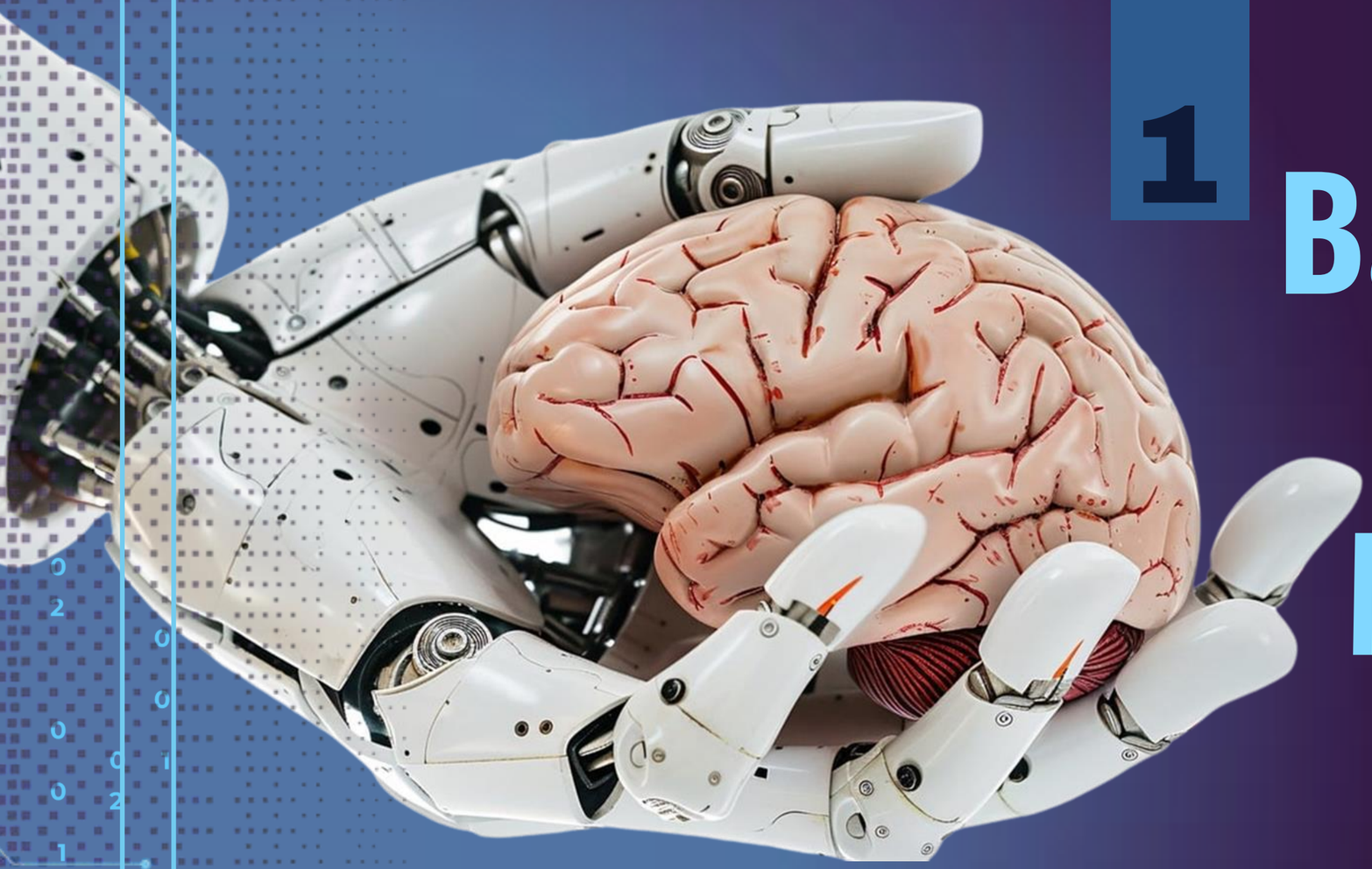
03. Proposed Methodology

04 Image Processing + Augmentation

05. DL Feature Extraction + ML Classification

06. Results

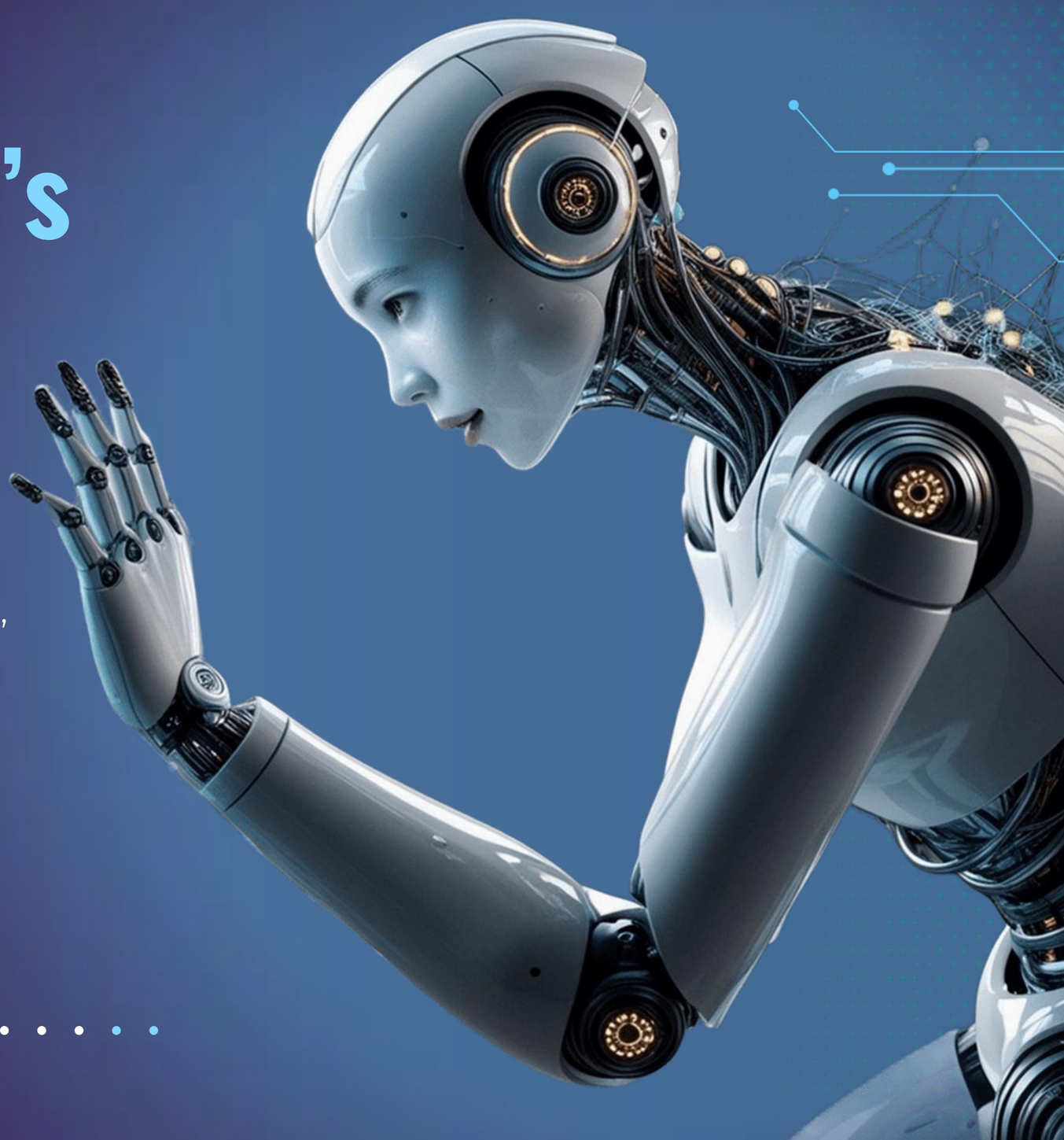
07. Conclusion & Future Work



1 Background & Motivation

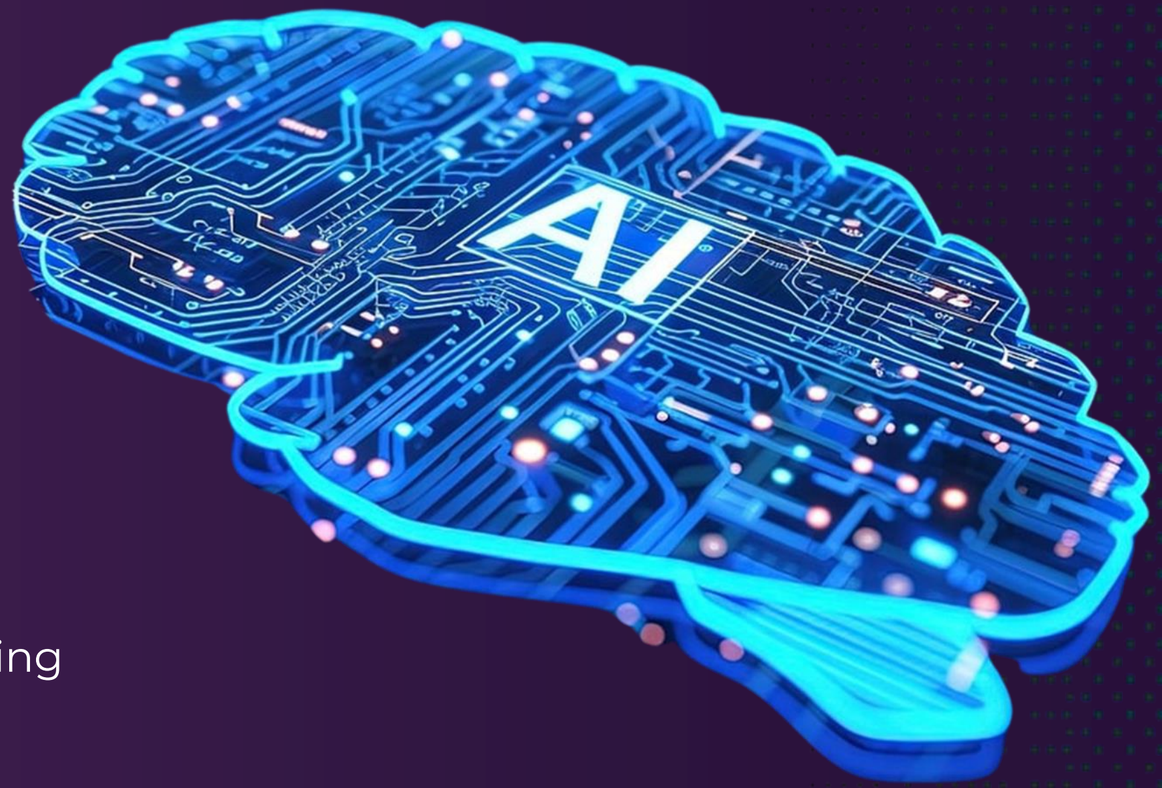
What is Parkinson's Disease?

- Parkinson's disease (PD) is a **progressive neurodegenerative disorder** affecting motor control and non-motor functions.
- Core motor signs include **bradykinesia**, tremor, and rigidity, used in clinical diagnostic criteria.
- PD also affects **speech/voice** (hypokinetic dysarthria), making speech a useful signal for detection.



Why Parkinson's & Why Speech?

- PD is a progressive neurodegenerative disorder; speech is commonly affected.
- Speech is **non-invasive, low-cost**, and easy to collect remotely.
- Goal: design a robust PD screening framework using speech-derived images.



Literature Review

Study	Reference (Year / Source)	Image/Audio Type	Dataset	Feature Extraction / Model	Best Performance (%)
LightAudioCNN for PD recognition	Springer **(2025)**	Mel-spectrogram / audio features	83 PD/HC participants & Italian dataset	Deep CNN (LightAudioCNN)	**~97.69%** (mean) (Springer Link)
Spectrogram-based multi-CNN feature fusion	*Frontiers Neurology (2025)*	Spectrogram images	61 PD + 70 HC	Pre-trained CNN fusion (MobileNetV3 + ShuffleNetV2)	**~95.56%** (Frontiers)
Voice-based early PD diagnosis (MDPI Sci Rep)	MDPI **(2025)**	Spectrogram / MFCC / raw audio	UAMS + mPower datasets	CNN (Inception V3), BiLSTM, CNN+GRU	**~97%** (BiLSTM) (MDPI)
Audio Spectrogram Transformers	*PubMed (2025)*	Spectrogram images	PC-GITA + Italian PD/HC	Audio Spectrogram Transformer (AST)	**~97.14%** (on one dataset) (PubMed)

Table : Summary of Image-Based Parkinson's Disease Detection Approaches

Research Gap & Contribution

Gap

- Many PD speech works focus on a **single representation** (only one feature type).
- Robustness under noise/recording variability remains challenging.

This thesis contributes

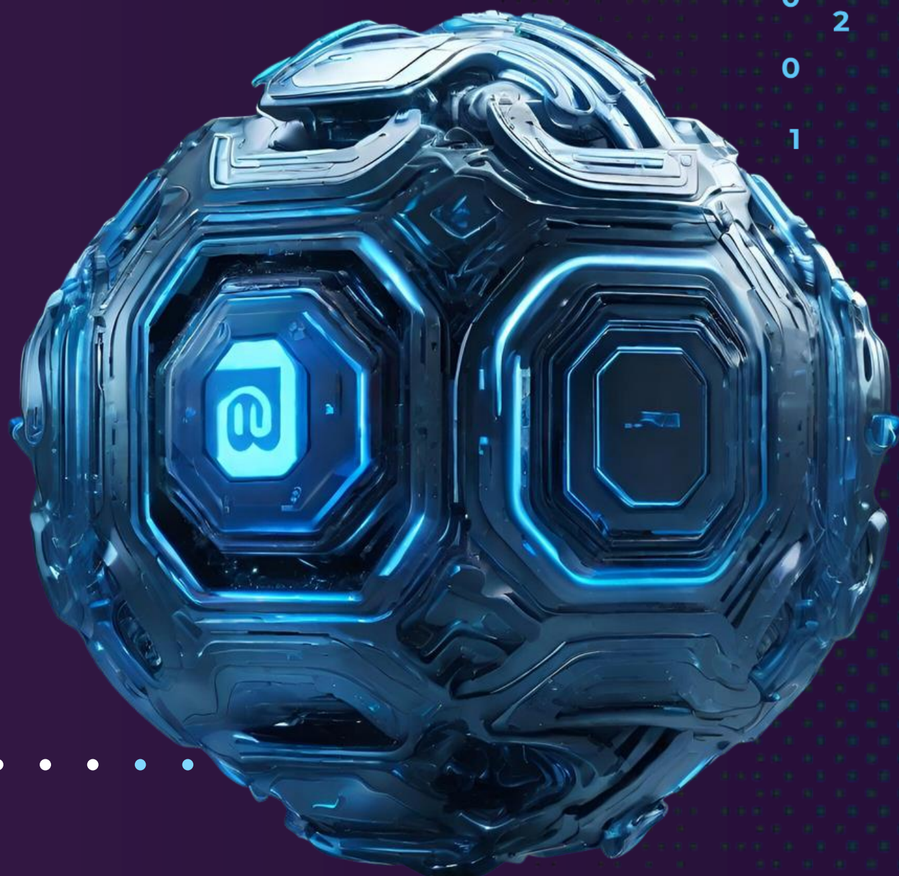
- Speech → **image-based representations**: Spectrogram + Formants (F1–F4) [15], [16]
- **Image preprocessing + augmentation** to improve robustness and balance. [56], [57], [60], [61]
- Systematic comparison of **10 DL models × 9 ML classifiers**.

Motivation

•In resource-constrained settings, awareness and access to specialist assessment can be limited.

•Cost and infrastructure barriers reduce feasibility of imaging-based support for large-scale screening.

•**Speech-based systems** can be low-cost, non-invasive, and scalable via common microphones/phones.





2

Dataset

Presents the dataset and summarizes the key recording conditions and participant distribution used in our experiments

Dataset: PC-GITA

PARTICIPANT DISTRIBUTION

100 speakers: **50 PD + 50 HC**, balanced gender (25M/25F each group).

RECORDING SETTINGS

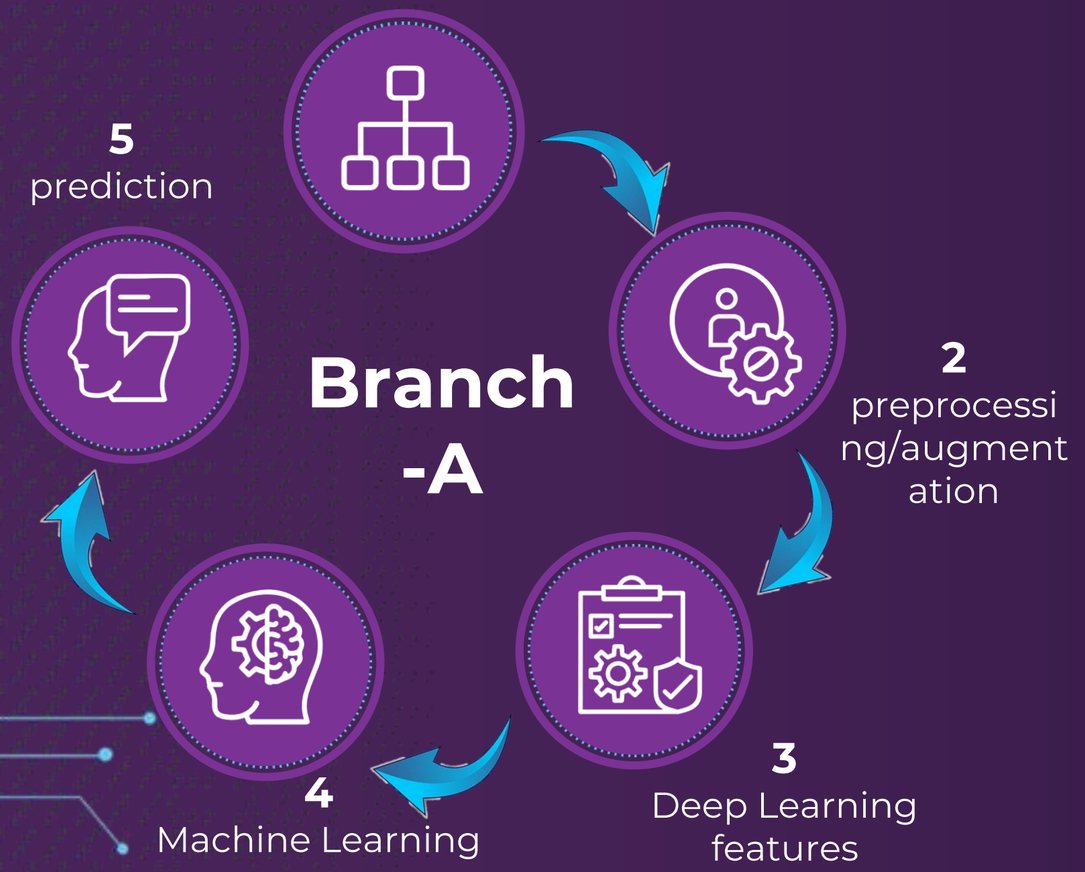
Recording in noise-controlled sound booth; **44.1 kHz**, **16-bit**

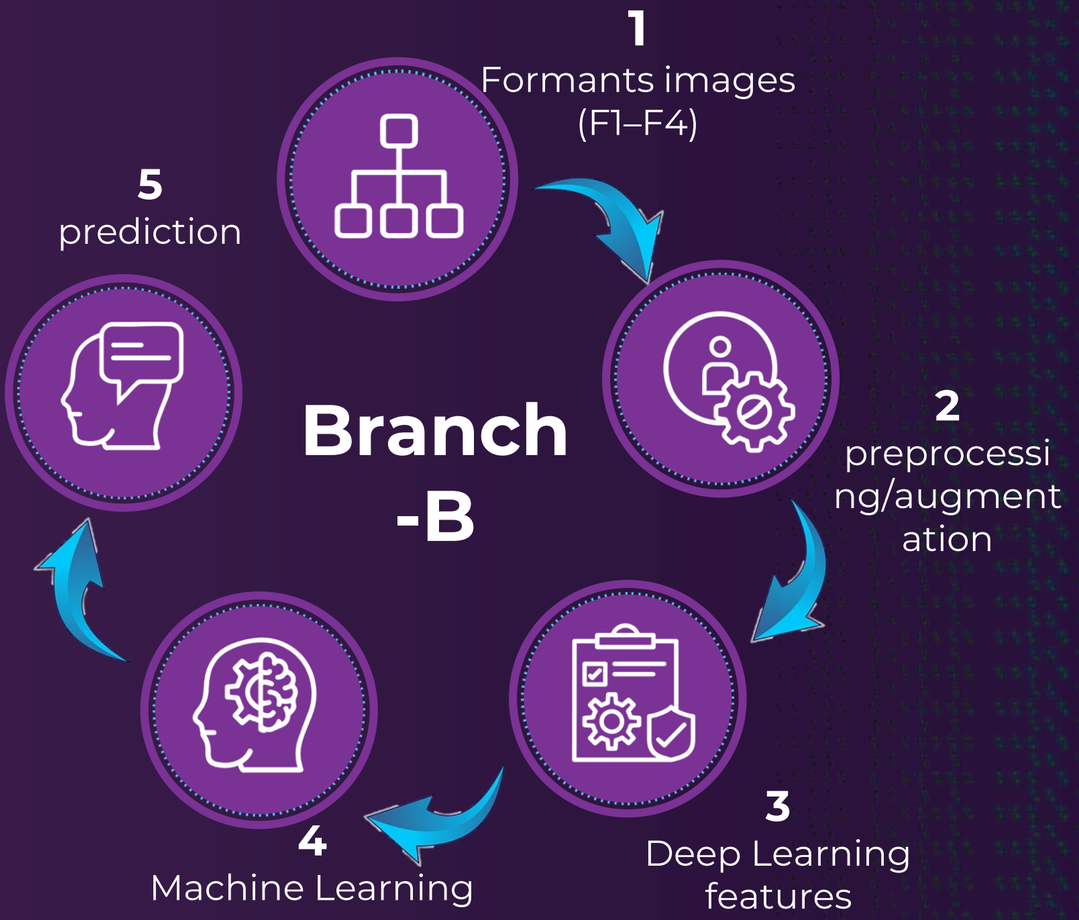
CLINICAL LABELING

PD labeled using UPDRS and H&Y; recorded in ON-state.

Main Two-Branch Pipeline

PC-GITA → 10-sec segmentation → **Branch-A**
PC-GITA → 10-sec segmentation → **Branch-B**





3

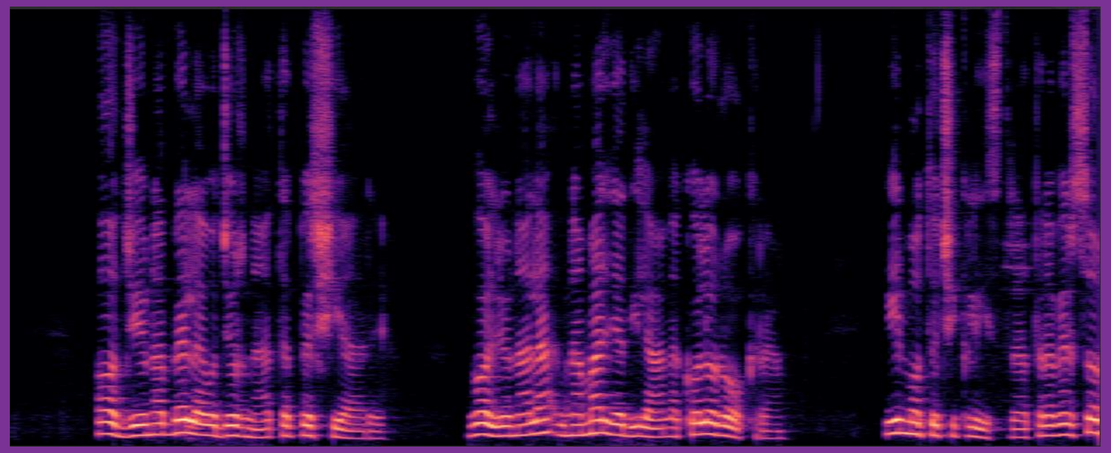
Image Processing + Augmentation



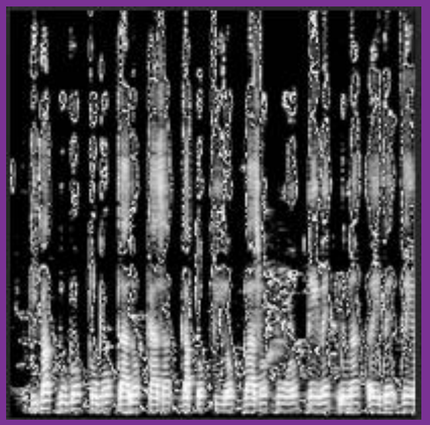
Branch-A: Spectrogram Preprocessing

Spectrogram preprocessing pipeline

- Grayscale → Median denoise → Edge emphasis
- Log compression → Min–max scaling → Contrast enhancement
- Resize 224×224 + normalization



Raw spectrogram vs processed spectrogram



Spectrogram Imbalance & Augmentation

Augmentation operations

Augmentation ops

- Rotation ±15°
- H-flip 0.5
- V-flip 0.5
- color jitter (0.2/0.2)
- Resized crop (0.8–1.0 → 224×224)

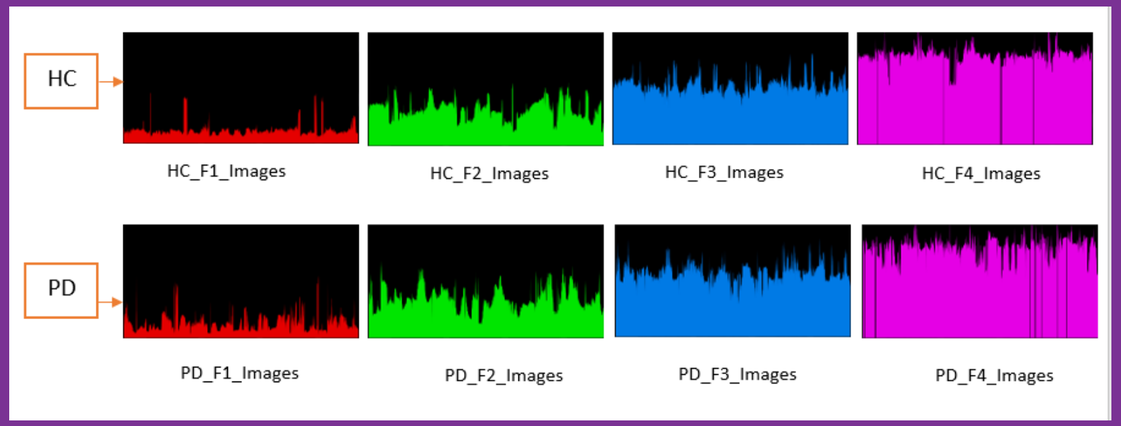
Before augmentation VS After augmentation

Class	Before Augmentation	After Augmentation
HC	162	1000
PD	298	1000
Total	460	2000

Branch-B: Formants (F1–F4) + Preprocessing

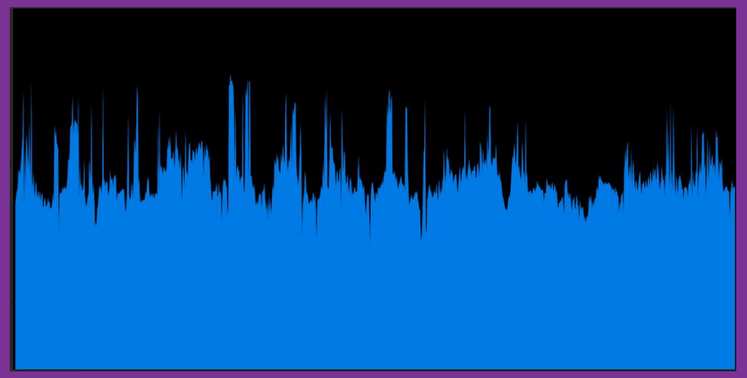
Formant representation

Extract formants and visualize as images (F1, F2, F3, F4)



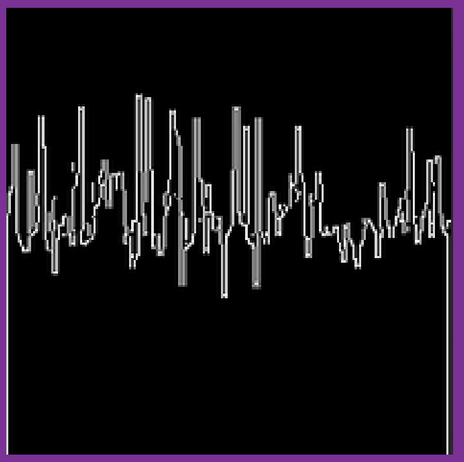
Formant preprocessing

Resize 224×224 → Grayscale → Normalize 0–255 → Gaussian blur → Canny edges



Raw Formants (F3)

processed Formants (F3)



Formants Imbalance & Augmentation

Augmentation operations

Augmentation ops

- Rotation ±15°
- H-flip 0.5
- V-flip 0.5
- color jitter (0.2/0.2)
- resized crop (0.8–1.0 → 224×224)

Before augmentation VS After augmentation

Formant Subset	HC(Before)	PD(Before)	HC(After)	PD(After)
F1	192	331	500	500
F2	192	331	500	500
F3	192	331	500	500
F4	192	331	500	500
Total (F1-F4)	768	1324	2000	2000

4

DL Feature Extraction + ML Classification



Deep Learning Models (Feature Extractors)

10 DL models used for feature extraction:

- AlexNet, VGG16, ResNet50, DenseNet121
- MobileNetV2, EfficientNetB0, Xception, InceptionV3
- ViT, CustomCNN

Key idea

- DL models are used to extract **feature vectors** from images
- Final decision is done by ML classifiers (next slide)

Machine Learning Classifiers



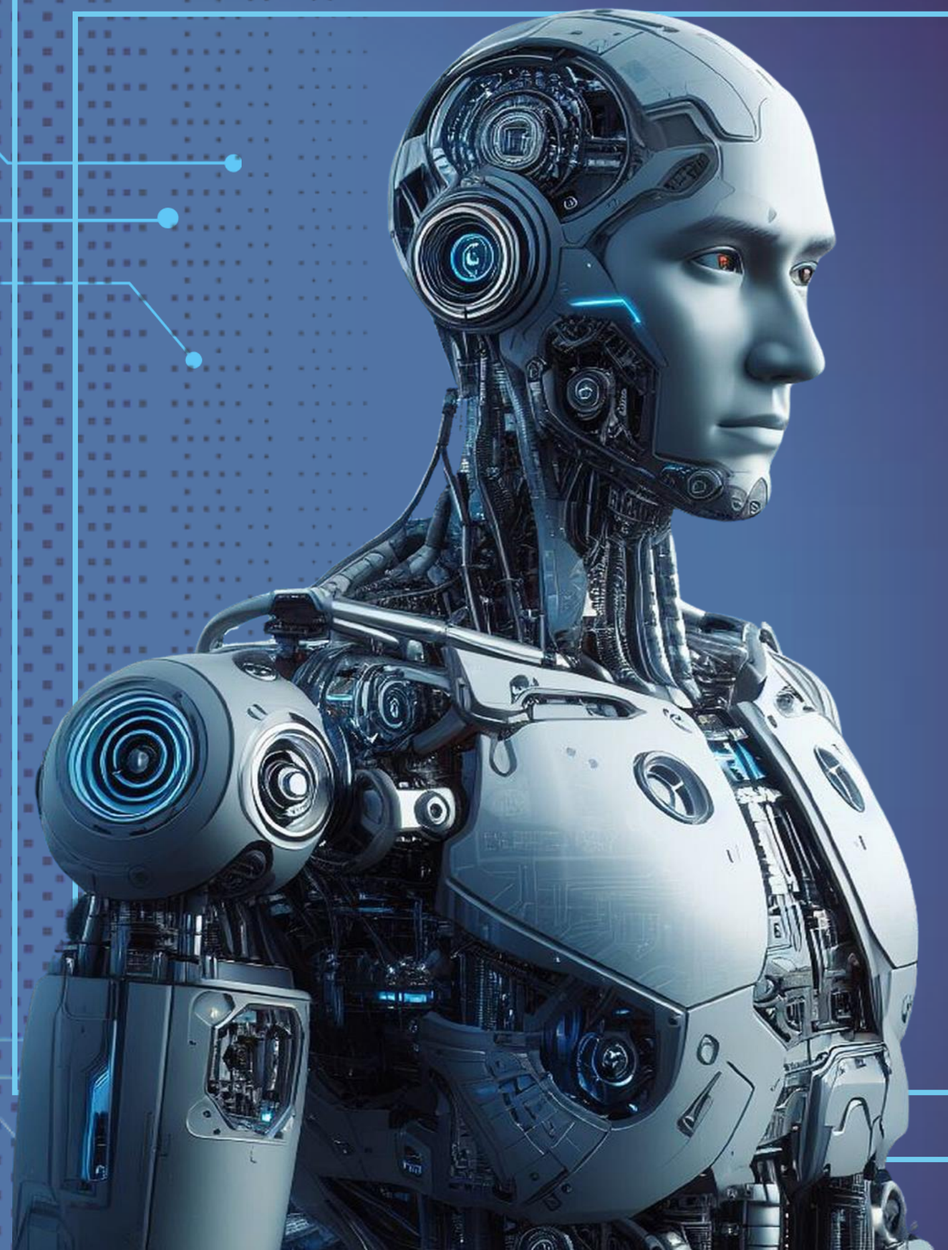
9 ML classifiers used on extracted deep features:

- SVM, Random Forest, KNN, Logistic Regression
- Decision Tree, Naive Bayes
- AdaBoost, Gradient Boosting, Extra Trees



Why ML after DL?

- Works well with deep embeddings
- More stable under limited biomedical data
- Fast training & comparison across many combinations



5

Proposed Methodology

Proposed Methodology Pipeline

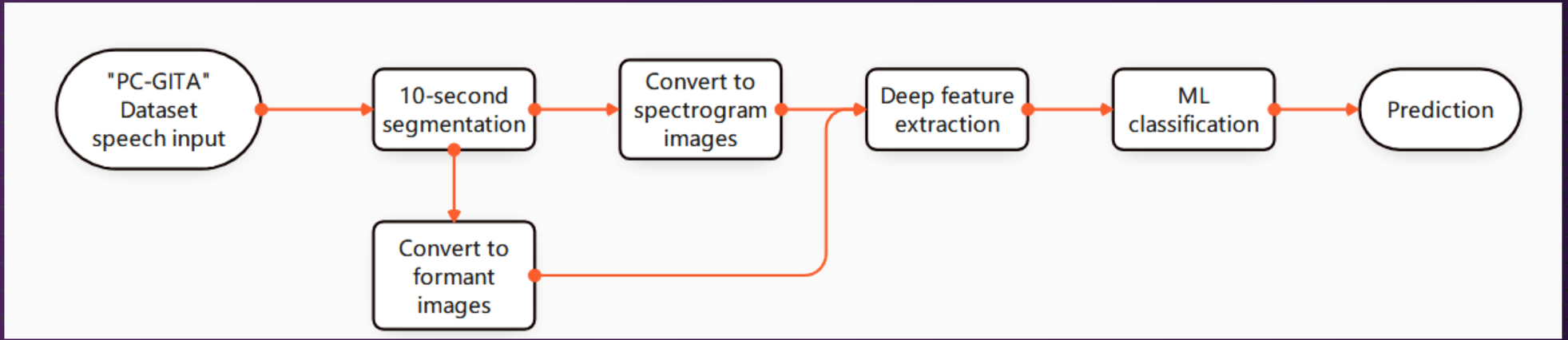


Figure : illustrates the overall workflow of the proposed system.

This figure summarizes our end-to-end workflow: PC-GITA speech recordings are segmented into 10-second clips, converted into spectrogram and formant images, transformed into deep feature embeddings, and finally classified using machine learning to predict PD vs HC.

Spectrogram Deep Feature Extraction and Classification

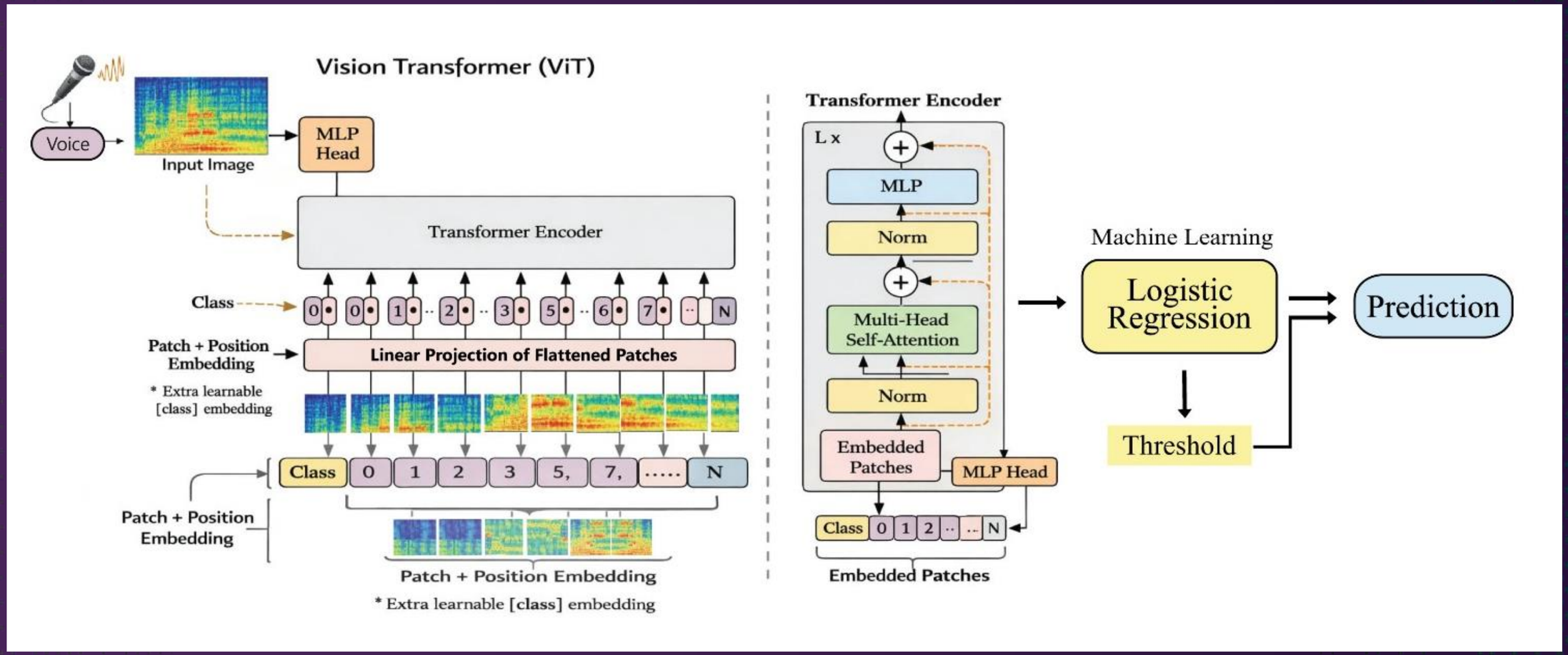


Figure : Architecture of the Vision Transformer–Based Feature Extraction and Logistic Regression Classifier

Formants Deep Feature Extraction and Classification

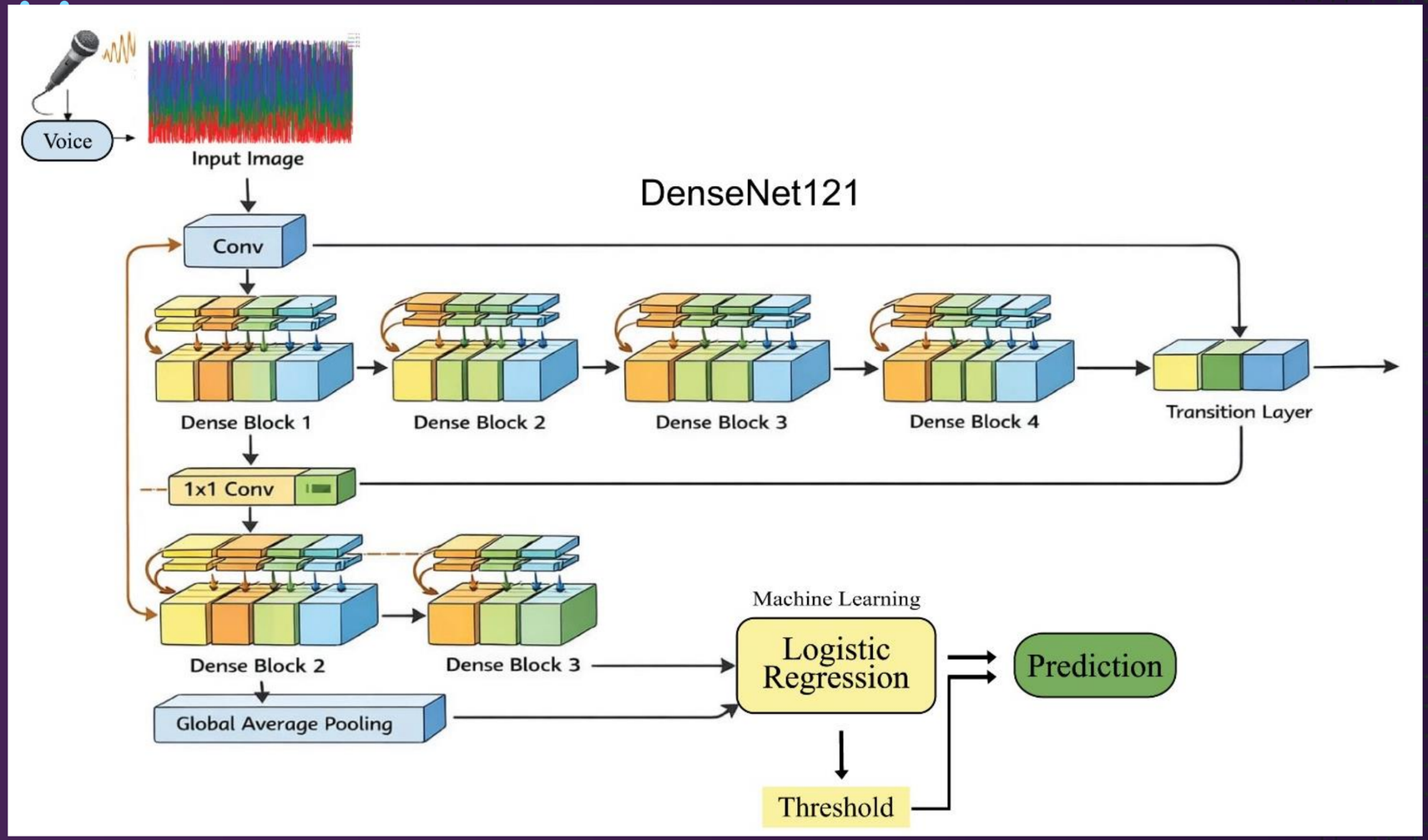


Figure 3.5 Architecture of the DenseNet121 Model with Logistic Regression Classifier



6

Results

Performance Analysis Using Spectrogram-Based Features

	Feature Extractor	Best Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Spectogram	AlexNet	Logistic Regression	93.50	92.80	93.00	92.90
	CustomCNN	Gradient Boosting	88.50	88.00	87.50	87.75
	DenseNet121	Logistic Regression	97.25	96.80	97.00	96.90
	EfficientNetB0	Logistic Regression	97.25	96.90	97.00	96.95
	InceptionV3	SVM	93.75	93.20	93.50	93.35
	MobileNetV2	Logistic Regression	96.00	95.50	95.75	95.63
	ResNet50	Logistic Regression	96.50	96.00	96.25	96.13
	VGG16	Logistic Regression	96.25	95.80	96.00	95.90
	ViT	Logistic Regression	99.00	97.04	98.50	97.77
	Xception	Logistic Regression	94.50	94.00	94.25	94.13

Table: Performance of Different CNN Feature Extractors Using Spectrogram Features



Performance Analysis Using Formants-Based Features

Feature Combination	Model	Classifier	Best Accuracy (%)	Precision	Recall	F1-Score
F1	AlexNet	Extra Trees	90.45	0.90	0.91	0.904
F2	DenseNet121	SVM	89.81	0.89	0.90	0.898
F3	VGG16	Logistic Regression	92.36	0.92	0.93	0.924
F4	VGG16	Logistic Regression	91.72	0.91	0.92	0.917
F1+F2	EfficientNetB0	AdaBoost	95.24	0.95	0.95	0.952
F2+F3	CustomCNN	SVM	95.24	0.95	0.95	0.952
F2+F4	MobileNetV2	SVM	95.24	0.95	0.95	0.952
F3+F4	VGG16	SVM	95.24	0.95	0.95	0.952
F1+F4	Xception	Logistic Regression	95.24	0.95	0.95	0.952
F1+F3	VGG16	Logistic Regression	95.24	0.95	0.95	0.952
F1+F2+F3	**DenseNet121**	**Logistic Regression**	**96.19**	**0.96**	**0.96**	**0.962**
F1+F3+F4	DenseNet121	Logistic Regression	96.19	0.96	0.96	0.962
F2+F3+F4	**AlexNet**	**Logistic Regression**	**96.19**	**0.96**	**0.96**	**0.962**
F1+F2+F4	InceptionV3	Logistic Regression	94.29	0.94	0.94	0.943
F1+F2+F3+F4	AlexNet	AdaBoost	95.24	0.95	0.95	0.952

Table: Performance of Different CNN Feature Extractors Using Formant Features

Comparison with Existing Spectrogram-Based Studies on the PC-GITA Dataset

Study	Spectrogram Type	Feature Extractor / Model	Classifier	Accuracy (%)
Little et al.[64]	Spectrogram	CNN	-	91.2
Wodzinski et al.[65]	Spectrogram	CNN	-	94.0
Orozco-Arroyave et al.[66]	Log Spectrogram	CNN	-	95.1
Vasquez-Correa et al.[67	Mel Spectrogram	CNN	SVM	95.8
Malekroodi et al.[68]	Multi-channel Spectrogram	Deep CNN	-	96.5
Gunduz[69]	Spectrogram	CNN + LSTM	-	96.9
Lee et al.[70]	Spectrogram	Audio Spectrogram Transformer	-	97.8
Proposed Method	**Spectrogram**	**ViT**	**Logistic Regression**	**99.0**

Table: Comparison of Spectrogram-Based Parkinson's Disease Detection Methods Using the PC-GITA Dataset

Spectogram & Formants Best Results Summary

Feature Type	Model	Classifier	Best (%)	Accuracy	Precision	Recall	F1-Score
Spectrogram	ViT	Logistic Regression	0.99		0.99	0.99	0.99
Formants	DenseNet121 + AlexNet	Logistic Regression	96.19		0.96	0.96	0.962

Table : Comparative Performance of Spectrogram and Formant-Based Features for Speech Classification

Future Work

CROSS-DATASET VALIDATION

Test on different datasets/languages/devices to check generalization.

EXPANDED SPEECH BIOMARKERS

Add jitter, shimmer, MFCC, and prosody features

REAL-TIME AND MOBILE DEPLOYMENT

Build a lightweight real-time/mobile version of the system.

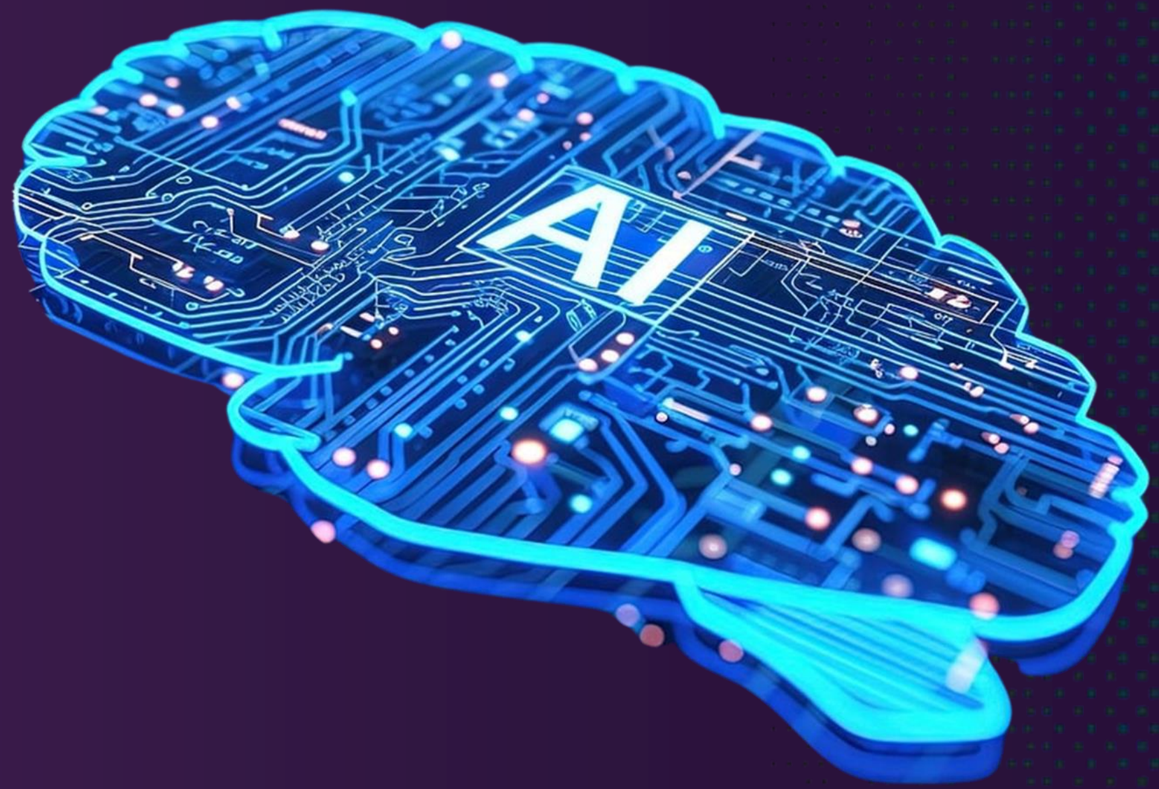
MODEL EXPLAINABILITY

Use Grad-CAM/attention maps to explain predictions



Conclusion

- Proposed **image-based PD detection** using spectrograms & formants.
- **Deep features + classical ML** achieved high accuracy.
- **Best results:** ViT + LR (spectrograms), InceptionV3 + LR (formants F1+F2+F3).
- Demonstrates **speech as a low-cost, non-invasive PD screening tool**





Thank You